

Sail geometry from astern — prior art, method and plan

Automating the three speed-team measurements that are made by hand in Rhino today: mast centreline → jib clew, mast centreline → jib leech at spreader 2, mast centreline → boom.

September 2026. Written against the 6 Sept speed-team material (`speedteam/notes-2026-09-06/01-12.jpg` , made from the 5 Sept sailing).

0. The one-paragraph answer

Nothing on the market does this. Every sail-vision product in sailing measures **shape** (camber, draft, twist) from a camera **on the boat**; the astern, off-boat measurement of **where the sail is relative to the boat** is done by hand in Rhino, SailTool or SailPack-Vision, everywhere, by everyone. The components to automate it all exist and are mature — they are just borrowed from 6-DoF object-pose estimation, not from sailing. The thing that decides whether this works is not the detector. It is **geometry**: at the ranges these photos are taken, one degree of being off the centreline puts ~140 mm of error into a 1,900 mm measurement, which is seven times the useful tolerance. The insight that makes the whole project tractable is that **the misalignment can be measured from the photograph itself, roughly twenty times more precisely than a photographer in a RIB can achieve it**. So the first generation should not *require* well-aligned pictures. It should *measure how badly aligned each picture is, and correct for it* — which is also what turns "a few good frames a day" into "every frame of a video".

1. What the measurement is today

From the 6 Sept compilations: pairs (sometimes triples) of stern shots, Northstar 76 beside Northstar 72 (and in one case Bella), each with the SSA instrument overlay burned in, and a red dimension line drawn horizontally across the picture at spreader-2 height with a number in millimetres — `1905` , `1929` , `1658` , `1782` , `1648` . That is *mast centreline → jib leech at spreader 2*, and the pairing makes it a direct A/B of two boats within 40 seconds of each other.

Measured off `01.jpg` : the `1905` dimension spans **696 px** on the 9,448 px-wide compilation. Each panel is a full-height crop of a 4000×6000 frame upscaled ≈2.26×, so on the **original** frame that dimension is ≈308 px, i.e.

≈ 6.2 mm per pixel.

The originals are Canon EOS R6 Mark II (6000×4000, 6.0 μm pixels) on an RF 100-500 mm, shot at 128–254 mm. A 6.2 mm/px scale at 254 mm implies a range of ≈260 m. Those two numbers — **6 mm/px, 260 m** — drive everything below.

Two facts worth noting before anything else:

- **The originals carry full EXIF; the files that reach SSA do not.** The overlaid/renamed exports in `photos/2026-09-05/` have lost `Make` , `Model` , `FocalLength` and `LensModel` — only `DateTimeOriginal` survives. Focal length is needed for the geometry, so either the pipeline preserves EXIF or focal length gets recovered from the image (it can be — see §4.3).
- **Measure on the original frame, never on the compilation.** The compilation is upscaled 2.26× and re-JPEGged; every edge in it has been resampled.

2. Prior art — sailing

2.1 On-board sail-vision systems (not applicable, but they set the bar)

System	Camera	Measures	Status
VSPARS (Univ. Auckland)	deck-mounted, looking up	stripe camber/draft/twist in 3D + rig deflection from coloured target dots	commercial; TP52s, Quantum Sail Scan
Onboard lidar	on the boat	camber / draft / twist at 25-50-75 % per sail, 4 Hz	already on Northstar 76 — see §3

System	Camera	Measures	Status
Masthead / forestay cameras	aloft, looking down	stripe shape upwind	the older standard; weight and windage aloft

VSPARS is worth one line only to say why it is *not* the reference here: it looks up from the deck at stripes, resolves rig target dots to better than 5 mm at the top of a 30 m rig, and its whole design problem is coping with extreme perspective on a sail seen nearly edge-on from below. None of that transfers to a long-lens frame shot from 260 m astern. Different optics, different problem. It is a useful accuracy benchmark and nothing else.

2.2 Off-boat digitising tools — the actual incumbents

These are what the speed team is really competing with, and what Rhino is standing in for:

Tool	What it is	Automation
The Sail Cloud (sailscan.thesailcloud.com)	cloud sail-scan; explicitly supports coach-boat photos ; compares mast bend and rake between boats on the water ; Mac + Windows; free trial then paid	"Auto-scan" for stripes; the off-boat rig work is manual digitising
SailTool (Curtin CMST)	Windows digitiser; draft, camber, offsets , twist, entry/exit, mast bend , leech twist, spinnaker flying distance	semi-manual: virtual protractor, tape measure, grid overlay. Freeware
SailPack-Vision (BSG / OneSails)	free image-measurement tool: load image, digitise stripes, measure distances and angles, superimpose images	manual
AccuMeasure (UK Sailmakers)	free; camber, draft position, twist per section	you send the picture in for analysis
Sailemetry	offline batch stripe analysis — 7,000 photos in ~30 min, no user input; needs contrasting draft stripes; zeroes twist against the spreaders	automated, but stripes only
SailWatcher (ISISLab, Salerno)	phone web app; user taps stripe positions	manual
North Sails SailScan app	phone export, one A4 page per sail	semi-automatic

SSA already parses two of these — `src/lib/sailScanParse.ts` reads North app and The Sail Cloud PDFs into `sail_scans`. So the ingest side of the stripe-shape world is done. What none of them do is the three numbers in the 6 Sept notes.

The closest anyone gets is **SailTool's "offsets" and "spinnaker flying distance"** and **The Sail Cloud's mast-bend/rake comparison between boats** — both of which are the same *kind* of measurement (a distance from a rig reference, digitised off a photo) and both of which are **manual clicks with an operator-set scale**. Nobody measures a jib clew position automatically, and nobody publishes an error budget for it.

2.3 Academic

- **Le Pelley & Modral, V-SPARS**, HPYD 2008 — the perspective-correction maths, and the 5 mm-at-30 m accuracy figure. vspars.com
- **Oliveira, "Sail and Rig Shape From Single Images"**, IST Lisboa 2016 — the right *framing* ("recovering 3D from a single 2D projection is under-constrained; exploit the fact that the geometry of the mast, sail stripes and leech is **partially known beforehand** to make it well-posed"), with poor results (20 %, 11 %, 4 % on three features) because it used a GoPro on the boom with no rig model. The idea is right; the execution had nothing to anchor to. We have the anchor.
- **Deparday et al.**, full-scale offwind flying shape with photogrammetry (*Ocean Engineering*, 2016) — four cameras on motorboats around the yacht; concluded that boats on moving spots are hard to synchronise and that RIB wake hampers the experiment. Reported <1.5 % accuracy. Relevant as the honest account of multi-boat photogrammetry's practical cost.
- **Maciel et al.**, monocular 3D sail shape with ArUco markers (*Machine Vision and Applications*, 2021) — single camera made well-posed by sticking known fiducials on the sail. We cannot stick markers on a rival, but we can on our own two boats (see §6, stage 4).

3. What SSA already has that none of them have

This is the part that changes the plan, and it is all in the repo already.

1. The boat is already instrumented with a lidar sail-shape system. The 4 Hz N76 export carries `JIB_CA_25` ... `JIB_TAL_75`, `MN_*`, `SPI_*`, `JIB_SAGx/y/z`, `JIB_LUFF%`, `JIB_LEECH%` plus the sailmaker's targets `T_JIB_*` — parsed in `src/lib/flatLogParse.ts`, filtered and tabled in `src/lib/lidarTables.ts`. **Do not rebuild camber/draft/twist from stern photos for our own boat; it is already measured at 4 Hz.**

2. The trim controls are logged. `JibIO%` (jib lead in/out), `Mainsheet`, `Vang`, `Forestay`, `UpDfclt%` / `LwDfclt%` (mast deflection), `ToeIn`, `KeelAngle`, `Heel`, `Trim` — at 4 Hz, on the same clock the photos are matched against. **This is a free, automatically-labelled validation set of essentially unlimited size:** every stern photo of our own boat has, at the same second, a sensor reading of the thing the photo is supposed to measure. No other project in §2 has that, and it is the difference between "we think it's accurate" and a published error figure.

3. Both compared boats are ours. Northstar 76 and Northstar 72 are the same programme (a team holds several boats). So the rig model — the thing that makes the single-image problem well-posed — is available for *both* sides of the comparison from day one. Rivals like Bella are a later, degraded mode.

4. The photo pipeline exists. EXIF-time → log-row matching, venue-local clock handling, instrument overlay rendering (`photoOverlay.js`, `PhotosTab.jsx`), cloud enrichment via `npm run media:upload`, and a `sail_scans` table with `photo_id`, `session_id`, `run_id`, `conditions`, `stripes`, `summary` JSONB that a new measurement family slots straight into.

5. Computer-vision foundations are already in the repo. `src/lib/sailscan-cv.ts` lazy-loads OpenCV.js and implements CLAHE, the gradient structure tensor and orientation masking; `sail-scan-ai/` is a vendored YOLO11 + SAM 2.1 stripe pipeline; `docs/sailscan/prior-art.md` is the stripe survey. The detector work is not starting from zero.

3.1 The rig dimensions already exist — for the whole fleet

An endorsed IRC certificate carries exactly the three things the photograph cannot supply, measured by a measurer, and it is issued for competitors too. Parsed from the six Maxi 72 certificates (`npm run irc:rigmodel -- <folder>`):

boat	sail no	LH	kg	P	E	J	HLU	HLP	rake	clew	leech	boom
Northstar III	GBR76X	23.20	17017	31.44	10.33	8.86	30.60	8.96	16.8°	-1.05	-0.35	-10.33
Jethou	GBR74R	23.50	15088	30.74	10.17	8.35	29.54	8.54	16.4°	-1.09	-0.34	-10.17
Bella Mente	USA45	22.55	14694	30.20	9.66	8.40	29.58	8.72	16.5°	-1.22	-0.38	-9.66
Balthasar	MLT5	21.95	15985	30.00	9.88	8.44	28.48	8.52	17.2°	-0.96	-0.27	-9.88
Jolt	GBR72N	21.95	15064	29.82	9.86	8.38	28.98	8.70	16.8°	-1.21	-0.36	-9.86
Django 7X	GBR8N	21.40	12147	28.52	9.63	8.53	28.12	8.83	17.7°	-1.16	-0.31	-9.63

Metres; the last three are fore-and-aft offsets from the mast, forward positive, and are *derived* from J/HLU/HLP rather than read off.

P is the scale reference, and it is the thing that was missing. It is not an athwartships length, but it does not need to be: seen from astern the mast lies in the plane perpendicular to the line of sight, so the mainsail hoist between the black bands projects at **full length**. Rake costs $\cos(\text{rake}) - 0.06\%$ at 2°, 0.2 % at 4° — and mast bend costs the arc-to-chord difference, about 0.02 % for 300 mm of sagitta over 31 m. At 31.44 m it is five times the baseline of a spreader and lands the scale an order of magnitude tighter: on a 6,000 px frame P spans ~5,000 px, so ± 1 px at each end is $\pm 0.04\%$.

J is the ψ baseline — forestay tack to the mast at deck, both centreplane points visible from astern — and **E puts the boom**. The jib's corners come out of J, HLU and HLP by construction.

Two caveats. J is measured to the *front* of the mast, not its centreline — half a section, ~200 mm, which scales ψ by the same fraction. And the clew's height up the luff is the one thing the certificate does not carry; it is taken as $15\% \pm 6\%$ of the luff, which is where the clew's ± 0.5 m comes from.

And the point that matters beyond our own boat: **five of those six are rivals**. Stage 4 had no answer for a boat whose rig model we do not have. IRC is the answer.

4. The physics that decides the project

This section is the reason to read the document. All numbers at $R = 260\text{ m}$, 6.2 mm/px , and a jib clew $\Delta d \approx 8\text{ m}$ forward of the mast (**substitute N76's real J** — the conclusions scale linearly with it).

4.1 Pixel noise is not the problem

At 6.2 mm/px , sub-pixel edge fitting ($\pm 0.3\text{ px}$ is routine for a high-contrast edge) gives $\pm 2\text{ mm}$ per endpoint. Against a $\pm 20\text{ mm}$ target, the detector has an order of magnitude in hand. **Anyone who frames this project as "we need a better detector" is solving the wrong problem.**

4.2 Misalignment is the problem — and it is measurable

Let ψ be the angle by which the camera sits off the boat's centreplane. A point Δd forward of the mast is displaced laterally in the image by $\Delta d \cdot \sin \psi$, and that displacement lands directly in the measurement:

ψ	error on a clew 8 m forward of the mast
1.0°	140 mm (7 % of a 1,900 mm reading)
0.5°	70 mm
0.2°	28 mm
0.1°	14 mm

To hold $\pm 20\text{ mm}$ by alignment alone the RIB would have to be within **0.6 m** of the extended centreline at 260 m — i.e. $\psi \leq 0.14^\circ$. That is not achievable, is not verifiable by eye, and is not what "mast and headstay in line" delivers in practice. **The current manual method has no way to know how far off it was.**

***Which target this bites is not where I first assumed.** The 8 m above was a guess. Northstar's IRC certificate (§3.1) gives J 8.86, HLU 30.60, HLP 8.96 — a 101 %-LP jib, and a 100 % jib's clew lands on the mast. Worked through, the clew sits about **1.1 m ABAFT** the mast and the leech at spreader 2 about **0.35 m** abaft it. So on the two jib targets ψ costs ~ 18 and $\sim 6\text{ mm per degree}$, not 140. The boom is the opposite case: E 10.33 m aft of the mast makes it **180 mm per degree**, and gives it a -4% depth error at 260 m . The principle is unchanged and the arithmetic is the same; what changed is which of the three measurements needs the care.*

But: any two identifiable points on the boat's **centreplane** at different fore-and-aft positions give ψ directly, because their lateral separation in the image is $\Delta d \cdot \sin \psi$. Candidate centreplane landmarks visible from astern: masthead, forestay tack / bow, gooseneck, traveller centre, backstay, transom centre, rudder stock. With a 8 m baseline and $\pm 1\text{ px}$ (6 mm) localisation of each,

ψ is recovered to $\pm 0.06^\circ$, i.e. $\pm 8\text{ mm}$ of residual error on the clew.

Use a 20 m baseline (bow to transom) and it is $\pm 0.025^\circ$. **The photograph measures its own misalignment about twenty times better than the photographer can control it.** This is the whole project in one sentence.

4.3 Perspective/depth is a systematic bias the manual method ignores

The camera is **astern**, so a point Δd forward of the mast is Δd farther away and is imaged **smaller** by the factor $R/(R+\Delta d)$. Scale it with the mast's mm-per-pixel and you **under-read** by $\Delta d/R$. At $R = 260\text{ m}$, $\Delta d = 8\text{ m}$ that is **-3.0% , i.e. $\approx 57\text{ mm}$ missing from a 1,900 mm reading**; at $R = 140\text{ m}$ (the 128 mm shots) it is -5.4% . The boom, being aft of the mast, is over-read by the same mechanism with the opposite sign.

A single uniform image scale — which is what a Rhino overlay gives you — is only valid in one plane. So today's numbers very likely carry a **few-percent range-dependent bias**. That is harmless when comparing two boats in one frame at the same range on the same day, and *not* harmless when comparing across days, across venues, or between two boats at noticeably different distances. **This is a testable claim** and should be tested early (§6, stage 0).

Correcting it needs R and Δd . R comes from the apparent size of any known rig dimension (spreader tip-to-tip, mast chord, P, J); Δd comes from the rig model. Both are free once the model exists. As a by-product, the same fit **recovers focal length**, which is why the stripped EXIF is survivable.

4.4 The definition question — world-horizontal vs boat-frame

The red dimension lines in the 6 Sept compilations are drawn **horizontal in the image**, on a boat heeled 21–25°. That measures a *world-horizontal* distance, not an *athwartships-in-boat-frame* one. Measuring horizontally to a mast that has leaned away gives **world-horizontal = boat-athwartships ÷ cos(heel)** — exactly, and independently of the target's height. At 23° heel that is ×1.086, and two boats 2° apart in heel read **1.6 % (≈31 mm) apart** for identical trim.

This is a definition, not a bug — but it must be pinned down before any code is written, and it is the single most important thing to decide (§7). The automated method can emit **both**, and heel is in the log for our boats.

Measured, 24 Sep — and it appears already decided. On 01.jpg panel 1 (N76, 11:53:00) the sea horizon runs at -23.20° to the frame (least squares over 71 samples, rms 1.1 px) while the mast stands within 0.64° of the frame's vertical (145 rows, rms 47 px — that residual is mast bend). So the panel has been rotated ~23° in PhotoScape to stand the mast up, and the red dimension line, horizontal in the rotated panel, is therefore perpendicular to the mast — the boat-frame athwartships measurement, not the world-horizontal one. The same numbers are a clean validation of the whole projection model: the mast's angle to the TRUE vertical is 23.20 – 0.64 = 22.56°, against a logged heel of 22.7°. Agreement to 0.14°, from a photograph, with no rig dimensions involved. Two consequences. SailTrim's "Across the mast" default already matches the historical Rhino numbers. And the rotation is invisible once applied — so a compilation panel must never be fed to the tool as if it were an original.

Camera roll, by contrast, is second-order (1° of roll moves the leech intersection ≈7 mm) and can be ignored once the pose is solved. On an unrotated original — 12:46:30, 5 Sept — the same method gives a camera roll of 4.8° and puts the mast 22.1° off true vertical against a logged heel of 22.1°. The camera on a RIB is never level; the horizon says by how much.

4.6 How far off the centreplane can this actually go?

§4.2 treats ψ as a small nuisance to be measured and removed. The real shots are not small: **the 4 Sept frames are 10–20° off**, which is a different regime, and the honest answer needed measuring rather than asserting. What follows is the tool's error against the full perspective camera in `src/lib/__tests__/support/rigCamera.ts` — ground truth, not self-consistency.

ψ	ψ recovered	jib leech (0.4 m aft)	clew (1.1 m aft)	boom (10.3 m aft)
0°	0.00°	0.00 %	0.00 %	0.00 %
5°	5.05°	-0.25 %	-0.30 %	-1.4 % (-13 mm)
10°	10.10°	-0.26 %	-0.36 %	-2.6 % (-23 mm)
15°	15.17°	-0.01 %	-0.15 %	-3.6 % (-32 mm)
20°	20.31°	+0.52 %	+0.33 %	-4.4 % (-40 mm)

The near targets are fine. Jib leech and clew stay inside ±1 % all the way to 20°, because they sit within a metre or so of the mast plane and ψ acts through depth. Those are the measurements the speed team asks for most, and they survive the angle.

The boom does not. At E = 10.33 m it loses 3–6 %, and the loss is a modelling residual rather than noise — it will not average away over frames. The reported sigma now carries it (±43 mm at 15°, against a -39 mm error), so the number stops claiming a precision it has not got, but the number itself is biased.

And one real bug this uncovered. An *athwartships* scale reference — the spreaders, and the steering wheels of §8 — images at $\cos \psi$ of its true extent, so the mm-per-pixel derived from it is too big by **sec ψ** : 3.5 % at 15°, 6.4 % at 20°. Worse, `solvePsi` inherits the same factor and returns **asin(tan ψ)**, so 20° reads as 21.4°. Both invert exactly and without iterating — `unbiasAthwartshipsScale` — and ψ then comes back inside 0.1° out to 25°.

A **vertical** reference has none of this: rotating the camera about a vertical axis does not foreshorten a vertical line. **P, between the mast's black bands, is the right scale for an off-centreplane shot**, and it is on every certificate.

What would fix the boom is not another correction term but a different model: the scaled-orthographic projection of §4.3 is what breaks at these angles, and replacing it with a full perspective solve is its own piece of work. It belongs with §8, which already needs the same machinery for parallax.

4.5 What about stereo off the RIB?

Not needed. Two cameras on a 2 m baseline at 260 m give depth to $\approx \pm 0.4$ m — useless for the transverse measurement (which is already good) but *just* enough for the §4.3 scale correction. The rig model supplies the same depth for free and better. Keep stereo in the back pocket for rival boats whose rig model we do not have.

5. Outside sailing — the methods worth stealing

Field	Technique	Why it matters here
Robotics / AR	Model-based 6-DoF pose (BOP benchmark; MegaPose, FoundationPose, render-and-compare)	Exactly our problem: known CAD model + one image $\rightarrow$ camera pose. Model-based localisation of <i>seen</i> objects improved >50 % (56.9 $\rightarrow$ 86.0 AR) on the BOP core sets since 2017. We have the "CAD model": the Rhino rig.
Classical CV	Single-view metrology (Criminisi, Reid & Zisserman)	3D affine measurement from one perspective view given a vanishing line and one reference length — the fallback when no rig model exists (rivals).
Aerial inspection	Thin-structure segmentation (CableNet, TTPLA, YOLO-based power-line detection)	Forestay, backstay, shrouds, leech: bounding boxes are hopeless for thin diagonal structures; pixel-level segmentation + Hough-style line fitting is the established answer.
Metrology	Sub-pixel edge localisation	Deep nets are good at <i>finding</i> edges and bad at <i>locating</i> them; the literature is explicit that classical sub-pixel methods still win on measurement accuracy. Use the net to find, classical maths to measure.
Wind energy	Blade tip-clearance from a single ground camera	Same shape of problem — a large flexible structure, one distant camera, millimetre-class deflection wanted — and the same conclusion: anchor to a known rigid geometry.
Broadcast	Bolt6 optical tracking, America's Cup	Millimetre-class asset positions recovered from TV feeds alone, in real time. Proof that model-anchored monocular tracking at this scale is production-grade, not research.

And the honest note from the AC world: *teams already run measurement programmes on chase-boat photographs to compare rigs and mast bend to millimetre accuracy, and treat it as a skill honed over years*. That is the craft this project is trying to turn into arithmetic.

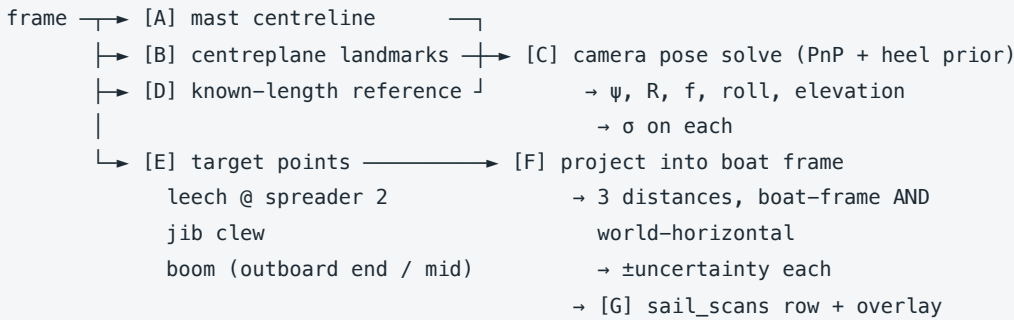
6. The method

Model-based, not pixel-based. Do not measure distances in the image and scale them. Solve for where the camera was, then measure in the boat's own coordinate frame. Same three numbers out, but with a stated uncertainty and no alignment requirement.

6.1 Inputs

1. **A rig model per boat** — mast section and centreline, spreader positions and lengths, hounds, gooseneck, boom length, forestay tack, transom centre, sheer line. Ten to twenty 3D points in boat coordinates is enough; the Rhino model already has them. Store as `boats.rig_model` JSONB (the `boats` table has no specs column today — this is the first thing it needs).
2. **The frame** (original, not compilation), with focal length if EXIF survived.
3. **The log row** at the photo's capture second — heel, trim, `JibI0%`, `Mainsheet`, `Forestay`, TWS/TWA. Already wired up.

6.2 Pipeline



- **[A] Mast centreline.** Two near-parallel high-contrast edges running the height of the frame; bisector gives the centreline, and the apparent chord gives a scale reference at the mast's depth. The structure-tensor machinery in `sailsan-cv.ts` already isolates near-vertical structure — the *inverse* of the mask it was written for.
- **[B] Centreplane landmarks.** Masthead, forestay tack, gooseneck, transom centre, backstay. Thin-structure segmentation for the stays; a keypoint model for the fittings.
- **[C] Pose.** Least-squares fit of the rig model's projection to the detected landmarks — 6 DoF, or 4 with focal length and heel taken from EXIF/log. Report **ψ as a first-class output on every measurement**; it is the quality score, and it replaces "was this picture well aligned?" with a number.
- **[D] Scale.** From the fit, not from an operator-drawn reference line.
- **[E] Targets.** Jib leech: a curve, intersected with the spreader-2 plane — so the answer is an intersection, not a keypoint, and is robust. Boom: a rigid known object, so its projected length gives its own depth as a check. Clew: the hard one — a single image constrains it to a ray, so its fore-and-aft position must come from the jib geometry (foot length from the tack) or from `JibI0%`. Its residual sensitivity is small once ψ is known (§4.2), but it deserves its own error bar.
- **[F] Output both frames**, with σ , and the pose parameters that produced them.
- **[G] Persist** as a `sail_scans` row (`source='ssa'`, a new `geometry` JSONB beside `stripes`), linked to `photo_id`, `session_id`, `run_id` — so it lands in Analytics, the report tables and Ask-the-data for free.

6.3 Video, later

With ψ measured per frame, **frame selection stops being about alignment**. The gate becomes: all landmarks visible, sharp enough, instruments steady (no manoeuvre), $|\psi|$ small enough that the correction is a correction and not an extrapolation — call it 3° . That passes far more frames than "mast and headstay in line" ever would, and the output stops being one number per photo and becomes **a time series of trim joined to the 4 Hz log**. That is the actual prize: not "what was the lead at 11:53", but "what was the lead through the whole beat, and what did the boat do about it".

Caveat for video: rolling shutter. A fast pan from a moving RIB skews the frame and the skew looks exactly like ψ . Either shoot stills bursts (the R6 II will do 40 fps electronic, same problem; 12 fps mechanical, no problem) or calibrate and correct the skew. **Test this before committing to video.**

7. Plan

Every stage ends with something usable, and every stage is validated against the logged sensors. Stages 0–4 are the near slice: three points on one sail, measured properly. **§8 is where that is going** — twist and draft at every stripe on both sails, for boats that have no instruments at all — and it reorders some of what follows.

Stage 0 — pin the definitions and kill Rhino (≈2 weeks)

The digitiser is built — Tools → SailTrim (`src/components/sailtrim/`, maths in `src/lib/sailTrim.ts`, detection in `src/lib/sailTrimCv.ts`, dimensions in `src/lib/rigModel.ts`). `/dev/sailtrim` opens it without a signed-in session. It takes an original frame, the mast edges at two heights, a scale reference, an optional centreplane baseline and the three targets, and returns each measurement **in both frames with a sigma**, beside the number the Rhino method would have given. ψ , the range and the camera roll are all reported. It recovers a synthetic target of known position to ± 5 mm through the full UI path.

It is also reachable from the photo itself — Photos → a photo → *Analyse sail geometry* opens the same digitiser on that photo's full-resolution original, and *Save to photo* writes the result back onto it: the three numbers on a card beside the instrument data, and, if the box is ticked, the lines burned into the picture beside the gauge overlay. That matters more than it sounds. A measurement that lives in a downloaded JSON is a measurement one person has; a measurement on the photo is one the whole team has, in the place they were already looking.

What travels is a small **annotation record** (`src/lib/sailTrimOverlay.ts`) — the geometry already resolved to pixels and millimetres, not the marks — carried in the photo row's `analysis_data`. The alternative, shipping the clicks and re-deriving on each device, would need the rig model to travel too and would quietly give a different answer the day a default changed. The record says which pixel frame its points are in, so the same three lines draw correctly on the 6000 px original, on the 480 px thumbnail and into a PDF.

What remains in this stage:

- **Decide the definition** (§4.4): boat-frame athwartships, world-horizontal, or both. Decide what "the clew" and "the leech at spreader 2" mean precisely — outer edge? centre of the ring? leech tape centreline? The tool computes both today so the decision is not blocking, but it is still the decision that matters most.
- **Feed it the rig model's real numbers** — **done, from the IRC certificates** (§3.1). Wire the heel straight off the log row rather than typing it in.
- **Find the frames the compilations were made from**. SSA holds 9 photos for 5 Sept — 11:42, 12:37×3, 12:46×3, 13:37, 14:25 — and *none* at 11:53, 11:58 or 12:03, which are the three moments the 6 Sept compilations use. Only the flattened, rotated, upscaled panels survived. Until those originals turn up, the "re-measure 20 Rhino dimensions" comparison has nothing to run on.
- **Re-measure 20 dimensions that were done in Rhino on 5–6 Sept** and compare. Two outcomes, both valuable: agreement (the tool is a drop-in replacement and Rhino is retired), or a systematic offset (§4.3's range bias is real and now quantified).
- *Ships*: the speed team stops using Rhino. *Produces*: the first labels.

Stage 1 — automatic pose, manual targets

Mostly built. Two of the four detections are automatic and validated on real frames; what remains is data, not code.

The horizon, found automatically (`detectHorizon`). The sea horizon is world-horizontal by definition, so it fixes the camera's roll exactly and makes the logged heel a *check* rather than an input. It runs the moment a frame is opened. On the six 5 Sept frames it fits to **0.9–1.3 px rms over 111–151 columns**.

Two things make it harder than it sounds, and both are in the code: a clear Mediterranean sky at the top of the frame is *bluer* than the sea just below the horizon (measured: 62–70 blue-minus-red against 38 for the pale sky above the water), so brightness relative to this frame's own sky is what separates them; and the boat's sails are navy — as blue and as dark as the water — so the search runs from the bottom up, the sea being the region that reaches the bottom edge. Everything that can go wrong pushes a column's answer *down*, so among candidate lines that fit, the highest one wins. That is what survives a headland filling more of the skyline than open water does.

The mast, from one click (`traceMastFromSeed`). The click says which edge; the trace follows it sub-pixel, up and down, as far as the contrast holds, and is drawn so the operator can see where it went. On the 5 Sept frames it covers **76–90 % of the frame height**. No network, no OpenCV — these are long, high-contrast, near-straight boundaries against plain sky, the one regime where classical edge finding plus a robust fit beats anything learned.

The end-to-end check, which needs no rig dimensions at all: the angle between the traced mast and the detected horizon *is* the heel. Against the log, over six frames:

frame	horizon	mast	heel from the photo	logged	Δ
12:37:46	−2.50°	20.76°	23.27°	23.5°	−0.23°
12:37:48	−1.86°	21.52°	23.38°	23.2°	+0.18°
12:37:52	−2.25°	23.95°	26.20°	24.8°	+1.40°
12:46:30	−4.71°	17.26°	21.97°	22.1°	−0.13°
12:46:31	−4.30°	16.54°	20.83°	22.4°	−1.57°
12:46:37	−4.57°	17.63°	22.21°	21.6°	+0.61°

Worst disagreement 1.57°, and the answers move by less than 0.2° between decoding at 900 px and 1600 px wide. Repeatable with `npm run sailtrim:validate -- <frames> --heel 23.5,23.2,...`, which is how any future change to the detectors gets checked.

Still to do, and all of it is the rig model rather than code:

- **The dimensions — now solved from IRC (§3.1).** `src/lib/ircCertificate.ts` reads a certificate into a rig model; paste one into the tool, or run `npm run irc:rigmodel -- <folder of PDFs> --out models` for the whole fleet. `src/lib/rigModel.ts` holds them per boat with the provenance of each — `designer`, `measured`, `derived` or `estimate` — and anything short of a real number propagates its sigma into every measurement. `supabase/migrations/0090` adds `boats.rig_model` and a `sail_trim` table; **not applied**.
- **Centreplane landmarks** are still clicked, not detected. Two points is a small ask, and ψ is the one number that most rewards being got right.
- **The log row** — heel, TWS, TWA, `JibI0%` — is still typed in rather than joined on the photo's capture second.

Stage 2 — automatic targets (≈6–8 weeks)

- Keypoint/segmentation model for leech, clew, boom, trained on Stage 0–1 labels plus synthetic renders of the rig model (the standard trick from the BOP world: render the CAD model under a distribution of poses and light).
- **Validate against the logged sensors:** predicted clew position vs `JibI0%`, predicted boom position vs `Mainsheet /traveller`, predicted leech vs `JIB_LEECH%` and `JIB_SAGy` from the lidar. This is the claim that sells the product, and the data for it already exists for every day we have sailed.
- *Ships*: upload a stern photo, get three numbers. The 6 Sept workflow becomes automatic.

Stage 3 — video and the time series (≈6 weeks)

- Rolling-shutter test first. Frame scoring, per-frame measurement, aggregation, join to the 4 Hz log, chart in Analytics.
- *Ships*: trim as a continuous channel rather than a handful of stills.

Stage 4 — boats we do not own (opportunistic)

- ~~No rig model for a rival.~~ — **solved by the certificates** (§3.1), and §8 argues this is the beginning of the interesting part rather than an opportunistic afterthought: the same certificates carry the sail girths, which are what turn an astern photograph into twist and camber.
- Expect a factor-of-2 worse uncertainty than our own boat and say so — the range is longer and the sail identification is inferred rather than known.
- Also: for our own boats, a handful of **passive markers** on the spreaders and boom (the Maciel 2021 trick) would cut Stage 1's uncertainty substantially for the cost of some tape. Worth a trial in Stage 1, not a dependency.

8. The longer game — twist and draft for boats with no instruments

Sections 6 and 7 measure three points on one sail. The destination is different and much larger: **twist and draft at every stripe, on the main and the jib, for a boat we do not own and cannot instrument**. Northstar has lidar and SailScan; the five rivals in §3.1 have a photograph and a certificate. If a photograph and a certificate are enough, SSA can say what a rival's sails are doing, which is not a thing anyone can currently buy.

The plan below is not a straight extension of Stage 4. The difficulty is distributed very differently from how it looks, and one quantity in the list is much harder than the rest.

8.1 What the astern projection preserves, and what it destroys

Take the camera dead astern, looking forward. It images the athwartships-vertical plane; fore-and-aft is the axis it collapses. Every quantity on the wish list sorts cleanly by which direction it mostly lives in:

quantity	mostly lies	from astern
leech offset at a given height	athwartships	measured directly — this is Stage 0
camber depth (the bulge)	athwartships	measured directly

quantity	mostly lies	from astern
twist (chord angle per stripe)	a ratio of the two	solvable, given the chord length
chord length per stripe	fore-and-aft	foreshortened to nearly nothing
draft position (% aft along the chord)	fore-and-aft	the hard one

The counter-intuitive line is the second. A sheeted mainsail's **chord** runs mostly fore-and-aft — Northstar's boom is E = 10.33 m aft of the mast while the leech is a metre or two to leeward — so the chord is precisely what the projection ruins. But the sail **bulges to leeward**, and leeward is athwartships: the belly is in the plane the camera resolves best. The thing that looks hardest to see from behind is in fact the easiest, and the humble length you would divide it by is the thing that is lost.

So camber % = (a quantity the photo measures well) ÷ (a quantity the photo cannot measure at all). Everything turns on where the denominator comes from.

8.2 The girths are the denominator, and the certificates carry them

They do — for both sails, for all six boats, parsed from the same PDFs as §3.1. MHW/MTW/MUW are the mainsail's widths at 1/2, 3/4 and 7/8 hoist; HHW/HTW/HUW are the headsail's at the same fractions of the luff:

boat	E	MHW	MTW	MUW	MHW/E	MTW/E	MUW/E	HLP	HHW	HTW	HUW	HHW/LP	HTW/LP	HUW/LP
Northstar III	10.33	7.04	4.93	3.63	0.682	0.477	0.351	8.96	4.90	2.66	1.48	0.547	0.297	0.165
Jethou	10.17	7.02	5.00	3.64	0.690	0.492	0.358	8.54	4.54	2.44	1.34	0.532	0.286	0.157
Bella Mente	9.66	6.74	4.91	3.74	0.698	0.508	0.387	8.72	4.60	2.50	1.40	0.528	0.287	0.161
Balthasar	9.88	6.78	4.91	3.71	0.686	0.497	0.376	8.52	4.56	2.51	1.40	0.535	0.295	0.164
Jolt	9.86	6.67	4.76	3.62	0.676	0.483	0.367	8.70	4.64	2.54	1.42	0.533	0.292	0.163
Django 7X	9.63	6.68	4.83	3.70	0.694	0.502	0.384	8.83	4.64	2.45	1.33	0.525	0.277	0.151

Two things follow, and the second is the surprise.

Twist and camber need no machine learning. Given the chord length at a stripe (the girth) and the athwartships component of the chord (the photograph), the fore-and-aft component is Pythagoras and the chord angle falls out; the difference between two stripes' chord angles is twist. Camber depth is the sagitta the photograph already shows, divided by the girth. For a rival's **mainsail** that is a complete answer from a photograph and a certificate, with no training data anywhere in it.

Normalised, the whole fleet is one sail. MHW/E spans 0.676–0.698 across six different designers — a **3.2 % spread**. HHW/LP spans 0.525–0.547, **4.1 %**. The planforms tighten as you come down the sail and loosen at the head, where designers actually differ: MUW/E spreads 9.8 %, HUW/LP 8.9 %. So for a Maxi 72 whose certificate we do not have at all, the girths are predictable from E or LP to a few per cent — and a few per cent on the denominator is a few per cent on the camber, which is inside the number's usefulness.

One gap worth naming: the widths are at 1/2, 3/4 and 7/8 — **there is no quarter-height width**. Draft stripes usually sit at about 1/4, 1/2 and 3/4, so the middle and upper stripes land on MHW and MTW almost exactly, and the lower stripe has to be interpolated between the foot (E, or HLP, at zero height) and MHW. That is a short, well-anchored interpolation rather than an extrapolation, which is why it is a footnote and not a problem.

8.3 The headsail gap is real, and SSA is the only one who can close it

The certificate dimensions **one** headsail. Northstar's is ID# J1.5-B, and **HLP** and **HSA** each appear exactly once in the file. The boat is rated to carry four. So J2, J3 and J4 — which is most of the wind range — have no girths at all, while the mainsail, which barely changes across the range, is fully described. The certificate covers the sail that changes least and omits the ones that change most.

Three ways in, best first:

1. **Measure them once on our own boat and keep them.** SailScan already measures chord per stripe. Every J2/J3/J4 scan is a girth table for a sail IRC never dimensioned — and these are fleet-standard sails from the same few lofts, so Northstar's J3 profile is a real prior for another Maxi 72's J3. This turns a hole in the certificate into a dataset SSA owns and nobody else has. It is the clearest case in this document of the instrumented boat paying for the un-instrumented ones.
2. **Scale from LP.** Smaller headsails are flatter, not scaled copies, so this is wrong — but wrong *systematically*, which means (1) measures the correction once and it applies thereafter.
3. **Which jib is up is already a Stage 0 output.** The clew position that shipped this week discriminates between J1/J2/J3 on its own, because the clew moves aft and inboard as LP shrinks. The sail identification the rest of this depends on is a by-product of what is already built.

8.4 Where the machine learning actually belongs

Not photograph → shape. **Learn the residual.**

Build the deterministic model of §8.2, run it on our own boat, and use lidar and SailScan to measure its *error*. Then fit a small correction — a handful of parameters — rather than a mapping from pixels to camber.

The reason is arithmetic, not taste. The scarce resource is not photographs and not lidar hours; it is their **intersection** — frames shot from the RIB while the lidar was running on a sail whose state is known. SSA holds nine photographs for 5 Sept. A residual model with a few parameters can be fitted on tens of paired samples and will tell you honestly when it is extrapolating. An end-to-end model needs thousands and, when it is wrong about a rival's camber, is wrong confidently and silently. That is the same argument that killed the sail-on-sail edge detector in Stage 0: a detector that picks the wrong edge with confidence is worse than no detector.

8.5 Video is not a convenience — it is the data engine

§6.3 and Stage 3 treat video as a later refinement that turns stills into a time series. That undersells it. A single pass down the transom at 30 fps is roughly **18,000 frames**, of which some hundreds will have a usable ψ and heel. Against nine photographs for a day, that is not an improvement in convenience; it is two orders of magnitude of training data for every stage that follows.

And the selection criterion already exists: ψ is measured, not assumed, so frames can be *scored and chosen* rather than hoped for. **Video frame selection should come before automatic target detection, not after it** — the detector wants labels, and every marked frame is a label, so the cheapest way to get a detector is to make marking cheap and frames plentiful first.

8.6 The ψ spread is parallax — geometry where the ML was planned

Draft position is the one quantity §8.1 marks as genuinely hard, because a position along a chord is a fore-and-aft measure and a single astern frame has no fore-and-aft information at all.

A burst of video frames does. Over a pass, ψ sweeps through several degrees, and those frames view the *same sail state* from different angles. That is parallax: the fore-and-aft shape one frame cannot see is constrained by the spread across many. A few degrees is a short baseline, but the sail is 10 m deep and the measurement is a shape rather than a range, which is a far kinder problem than the stereo depth estimate §4.5 dismissed.

This is the most valuable untested idea in this document, because it converts the hardest sub-problem from one needing training data into one needing none. It should be tried on a single video before any commitment is made to the learned route.

8.7 Already measured, currently thrown away: mast bend

`traceMastFromSeed` follows the mast sub-pixel over 76–90 % of frame height and then **fits a straight line through it and discards the curve**. Those points are a mast-bend measurement — free, on every frame, for rivals as well as for us, and bend is what sets the mainsail's entry. It should be kept and reported rather than fitted away. The fix is small and the data is already in hand on every frame ever marked.

8.8 The order this argues for

Each step reuses what the step before it built, and none of them is blocked on the one after:

- 1. **More targets, same geometry.** Leech at every spreader and every stripe, main *and* jib. [leechTargets](#) and the reference-height machinery generalise from one height to N almost unchanged. Highest value per unit of work in the whole document, and it needs no new science.
- 2. **Twist and camber from the girths** (§8.2), validated against our own lidar and SailScan. This either works or says exactly what is wrong.
- 3. **Video frame selection** (§8.5) — the data engine, before the detector.
- 4. **Automatic targets**, trained on the labels steps 1–3 have been generating all along.
- 5. **Parallax for draft position** (§8.6); the residual correction (§8.4) only for what parallax and geometry leave over.
- 6. **Rivals.** By this point the only thing that is theirs rather than ours is the certificate, and §3.1 has six of those.

The order matters more than any single step in it. Steps 3 and 4 are routinely run the other way round, which is how projects like this end up hand-labelling for months to train a detector on a corpus that a week of video would have made redundant.

9. Risks, honestly

Risk	Severity	Mitigation
The definition (§4.4) is settled late and Stages 0–2 measure the wrong thing	high	Settle it in Stage 0, before code
Rig model unavailable or inaccurate for N72	high	It gates everything from Stage 1. Confirm it exists <i>this week</i>
Clew depth ambiguity	medium	Constrain from jib foot + JibI0% ; publish its wider error bar; do not pretend it is as good as the leech measurement
Rolling shutter kills the video stage	medium	Test in Stage 3 before building on it
EXIF stripped upstream	low	Recovered by the pose fit (§4.3); also fix the export so it stops happening
Motion blur / spray / backlit sails	low	Frame scoring rejects them; at 1/1250 s there is little blur
Detector accuracy	low	§4.1 — there is an order of magnitude in hand

The thing that kills this project is not vision. It is measuring something subtly different from what the speed team means, very precisely.

10. What I need decided

- 1. **The definition.** Probably already answered — §4.4 measures the 6 Sept panels as rotated ~23° to stand the mast up, which makes the existing numbers **boat-frame athwartships**. Confirm that is deliberate rather than an artefact of making the panels look tidy side by side, and SailTrim's default is already right.
- 2. **The exact target points.** Clew: ring centre or sail corner? Leech at spreader 2: at the spreader *tip* height or the spreader *root* height on the mast? Boom: outboard end, or a fixed station along it?
- 3. **The tolerance that matters.** Is ±20 mm right, or is ±50 mm plenty, or does the team actually need ±10 mm? The whole error budget is set by this answer. With P as the scale and ψ measured, the jib targets should now sit inside ±20 mm; the boom is the one that will need the baseline marked every time.
- 4. **~~Does the Rhino rig model exist for both N76 and N72?~~** — moot. The IRC certificates carry everything the measurement needs (§3.1), for the whole fleet, measured and endorsed. A rig drawing would still improve two things: the mast's athwartships section (a better short-range scale check) and the height of spreader 2 (which is what the leech target is defined against, and is currently taken as 20 m).
- 5. **Is the 5–6 Sept Rhino session reproducible** — the same operator measuring the same dimensions again — so Stage 0 has a human-repeatability figure to compare against? Without it we can say the tool is consistent, but not that it beats a person.

Sources

Sailing: [VSPARS](#) · [Le Pelley & Modral, V-SPARS, HPYD 2008](#) · [The Sail Cloud](#) · [SailTool, Curtin CMST](#) · [SailPack-Vision](#) · [Sailemetry](#) · [AccuMeasure, UK Sailmakers](#) · [SailWatcher](#) · [Oliveira, Sail and Rig Shape From Single Images, IST 2016](#) · [Deparday et al., full-scale offwind flying shape](#) · [Maciel et al., monocular 3D sail shape with passive markers](#) · [How the TP52 fleet uses America's Cup tech](#)

Computer vision: [BOP Challenge 2024](#) · [BOP benchmark](#) · [MegaPose](#) · [FoundationPose](#) · [Criminisi, Reid & Zisserman, Single View Metrology](#) · [TTPLA power-line dataset](#) · [Vision-based power line and pylon detection](#) · [Sub-pixel edge localisation](#) · [Wind-turbine blade tip clearance by machine vision](#)

In-repo: `docs/sailscan/prior-art.md`, `src/lib/sailscan-cv.ts`, `src/lib/flatLogParse.ts`, `src/lib/lidarTables.ts`, `src/lib/sailScanParse.ts`, `supabase/migrations/0035_boat_config_sails.sql`, `sail-scan-ai/`.